Fourteen new weights of the Reed–Muller code RM(7, 14)*

Horace Dong
Independent researcher
maxwelldhx@gmail.com
ORCID: 0009-0002-7554-7548

October 2026 (version 2)

Abstract

The binary Reed–Muller code RM(7, 14) has minimum distance $d = 128$. Leuenberger and Albrizzio (arXiv:2606.21425) showed that every even number from 312 to $2^{14} - 312$ is a weight of RM(7, 14), except possibly 22 values. We show that 14 of them are weights, by explicit codewords. We also give explicit codewords for all even numbers from 320 to $2^{14} - 320$ except 322, 326, 330, 334 and their complements. Hence, for the middle-interval formulation of an open question of Carlet used here, at $(c, m) = (7, 14)$ every even number in $[320, 16064]$ is a weight if and only if 322, 326, 330 and 334 are weights. For them we prove non-existence in restricted families and reduce each of them to a question about RM(6, 13) and RM(7, 13). We also give a proof of a result of Lou and Wang: their Conjecture 2, which would imply this middle-interval assertion, is false for $m = 6$. Every result of the paper is formally verified in Lean 4. The results were found and formalised by AI coding agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

For $0 \le r \le m$, the binary Reed–Muller code $\mathrm{RM}(r, m)$ consists of the truth tables on $\mathbb{F}_2^m$ of the polynomials in $x_1, \dots, x_m$ over $\mathbb{F}_2$ of degree at most r, where $x_i^2 = x_i$. We identify a codeword with its polynomial (its algebraic normal form), write $\mathrm{wt}\, f$ for the number of points at which $f = 1$, and call the set of weights of the codewords the *weight spectrum*. The minimum distance of $\mathrm{RM}(r, m)$ is $d = 2^{m-r}$. Kasami and Tokura [5] determined the weights below $2d$, and Kasami, Tokura and Azumi [6] those below $2.5d$. The whole weight spectrum of $\mathrm{RM}(m-c, m)$ is known, for all $m \ge 2c$, for $c \le 2$ (classical), for $c = 3, 4$ (Carlet and Solé [3]), for $c = 5$ (Carlet [2]) and for $c = 6$ (Lou and Wang [8]). Carlet asked [2] whether, for every $c \ge 1$ and $m \ge 2c$, the weight spectrum of $\mathrm{RM}(m - c, m)$ consists of 0, 2^m, the weights in $[2^c, 2^{c+1}]$ given by Kasami and Tokura and in $[2^{c+1}, 2^{c+1} + 2^{c-1}]$ given by Kasami, Tokura and Azumi, their complements to 2^m, and *all* even numbers in $[2^{c+1} + 2^{c-1}, 2^m - 2^{c+1} - 2^{c-1}]$. (We paraphrase the wording of Carlet's extended abstract [1] of [2], which is also that of [7, Conjecture 1]; we could not access the full text of [2]. In the earlier conjecture of Carlet and Solé [3, Section 6] the last interval starts at $3 \cdot 2^c$.) Since $d = 2^c$, the last interval starts at $2.5d$. The first open case is $(c, m) = (7, 14)$: here $d = 128$, and the last interval is $[320, 2^{14} - 320] = [320, 16064]$.

* *Version 2.* Every result of this version is formally verified in Lean 4 (Section 7); version 1 formalised only Theorem A. Version 1, titled "The weight spectrum of the Reed–Muller code RM(7, 14) up to four weights", also determined the weights below 320, using a theorem of Kasami, Tokura and Azumi that we could not consult and have not formalised; that determination and every other claim resting on the theorems of Kasami and Tokura or of Kasami, Tokura and Azumi have been deleted, among them Proposition C′ and the extension of Theorem C to arbitrary g_1, g_2. Theorem C was proved earlier by Lou and Wang [9]. The computations and heuristic searches of version 1 are no longer reported. The main changes are listed at the end of Section 7.

Leuenberger and Albrizzio [7, Proposition 4] showed that every even number from 312 to $2^{14} - 312$ is a weight of RM$(7, 14)$, except possibly the 22 elements of

$$M = \{322, 326, 330, 334, 354, 8174, 8178, 8182, 8186, 8188, 8190,\\ 8194, 8196, 8198, 8202, 8206, 8210, 16030, 16050, 16054, 16058, 16062\};$$

parts of their proof rest on computer searches whose details are not given (Lemmas 6 and 9 of [7]). They were unsure whether this is the whole spectrum, and remarked that their constructions would produce all of M from the eight weights $322, 326, 330, 334, 354, 4378, 4380, 4382$. Their constructions follow Lou and Wang, whose Conjecture 2 [8] would give 322, 326, 330, 334 and 354 (see Theorem C).

Results. Throughout, we write

$$U_0 = \{322, 326, 330, 334\}, \qquad U = U_0 \cup (2^{14} - U_0) = U_0 \cup \{16050, 16054, 16058, 16062\},$$
$$K = \{0, 128, 192, 224, 240, 248, 252, 254, 256, 272, 288, 296, 304, 308, 312, 314, 316, 318\}.$$

The set K consists of the numbers below 320 that [7, Proposition 4] asserts to be weights of RM$(7, 14)$. Its elements are weights (Theorem B). Version 1 of this paper also asserted, citing Kasami, Tokura and Azumi [6], that they are all the weights below 320; we could not consult [6], and we make no claim about other numbers below 320.

Theorem A. *The following* 17 *numbers are weights of* RM$(7, 14)$*:* 354*,* 4378*,* 4380*,* 4382*,* 8174*,* 8178*,* 8182*,* 8186*,* 8188*,* 8190*,* 8194*,* 8196*,* 8198*,* 8202*,* 8206*,* 8210 *and* 16030*. Table 1 gives codewords with at most seven monomials.*

Fourteen of these numbers, the elements of $M \setminus U$, are listed in [7] as possibly missing; to our knowledge they are new. The other three, 4378, 4380 and 4382, are not new: they are weights by [7, Proposition 4], and the remark of [7] lists them among the eight weights from which its constructions would produce M. We include codewords for them because we do not use [7, Proposition 4].

Theorem B. *Let* $S_0 = \big(K \cup \{320, 322, \ldots, 16064\} \cup (2^{14} - K)\big) \setminus U$*. Every element of* S_0 *is the weight of an explicit codeword of* RM$(7, 14)$ *with at most eight monomials.*

In Carlet's question the weights below $2.5d$ and their complements are, by definition, those given by [5, 6], and all weights are even by (P1) in Section 2. So we read the question, for $(c, m) = (7, 14)$, as asking whether every even number from 320 to 16064 is a weight.

Corollary. *For* $(c, m) = (7, 14)$*, Carlet's question in this reading has a positive answer if and only if* 322*,* 326*,* 330 *and* 334 *are weights of* RM$(7, 14)$*.*

Proof. If every even number from 320 to 16064 is a weight, so are these four. Conversely, if they are weights, so are their complements, by (P1); with Theorem B, every even number from 320 to 16064 is a weight, since those not in S_0 are the elements of U. □

Construction 2 of [8] takes $g_1 = x_1 \cdots x_m + x_{i_1} \cdots x_{i_m} + x_{i_{m+1}} \cdots x_{i_{2m}}$ and $g_2 = x_1 \cdots x_m + x_{i_{2m+1}} \cdots x_{i_{3m}} + x_{i_{3m+1}} \cdots x_{i_{4m}}$ with $1 \le i_j \le 2m$, and $a_1, a_2 \in \mathbb{F}_2$, and forms the concatenation $g = 0 \parallel (g_1 + a_1) \parallel g_2 \parallel (g_1 + g_2 + a_2) \in$ RM$(m + 1, 2m + 2)$ of four truth tables (Section 4). Conjecture 2 of [8] states that, for $m \ge 4$, the weights of these codewords include $2^{m+2} + 2^m + 2i$ and $2^{2m} + 2^m + 2i$ for $0 \le i < 2^m$; for $m = 6$ the first family is $\{320, 322, \ldots, 446\}$, which contains 322, 326, 330, 334 and 354.

Theorem C. *Conjecture 2 of [8] is false for* $m = 6$*: no codeword of Construction 2 with* $m = 6$ *has weight* 322*.*

Lou and Wang proved this first [9, Proposition 2]; we had not seen their note when version 1 of this paper was released. We include Theorem C with a proof that is formalised here. Version 1 also stated it for arbitrary $g_1, g_2 \in \mathrm{RM}(6,12)$; that extension used the weights of $\mathrm{RM}(6,12)$ below 160, which [8, Lemma 2] takes from [5, 6], and it is not part of version 2. (In [9] the construction is called Construction 1 and our m is their d; we follow the numbering of arXiv:2406.03803v1.)

No sum of at most three indicators of flats of dimension at least 7 (Proposition D) and no sum of at most five monomials of degree at most 7 (Proposition E) has a weight in U. Proposition F reduces each $w \in U_0$ to a question about $\mathrm{RM}(6,13)$ and $\mathrm{RM}(7,13)$. Every result of the paper is formally verified in Lean 4 (Section 7).

Prior work. We found no earlier claim that an element of $M \setminus U$ is a weight of $\mathrm{RM}(7,14)$. We searched on 26 September 2026, 06:41–07:15 and 13:43–14:03 UTC: the arXiv (listing of [7], still at version 1 at 13:43 UTC, and API searches); the works citing [7, 8, 2, 3] in Semantic Scholar; Crossref; zbMATH Open; the MathSciNet reference lookup; MathDB, whose entries on Carlet's question and on Conjecture 2 of [8] were open, without solutions or comments; MathOverflow and Mathematics Stack Exchange; the IACR ePrint archive; HAL; GitHub, including `trureturing` [13] and other repositories of AI-assisted work on open problems; and web search. Citation queries to OpenAlex and keyword searches of Semantic Scholar hit rate limits; the full texts of [6], [2] and the published version of [8] were not accessible, so we quote [8] and [7] from their arXiv versions 1; we did not search Google Scholar, and Scopus was not accessible. A final check on 26 September 2026 between 15:49 and 15:52 UTC (the arXiv, where [7] still had only version 1; GitHub; MathDB; and MathOverflow) found nothing new. At that time we could not read the note [9] of Lou and Wang (online 4 April 2026). We read it after version 1 was released: it proves that Conjecture 2 of [8] fails for $m = 6$, with the weight 322, which is our Theorem C [9, Proposition 2]; it studies the codewords $g_1 \parallel g_2 \parallel g_3 \parallel g_4$ of weight 322 with $g_i \in \mathrm{RM}(6,12)$; and it asks whether 322 is a weight of $\mathrm{RM}(7,14)$ [9, Open problem 1]. Before the release of version 2 we repeated the searches on 9 October 2026 between 00:37 and 01:17 UTC: the arXiv, where [7] and the arXiv version of [8] still had only version 1, including all new submissions and replacements in cs.IT and math.CO announced up to 8 October 2026; the works citing [7, 8, 9, 2] in Semantic Scholar and OpenAlex, and those citing [8, 9, 2] in zbMATH Open, which does not list [7]; the citation counts in Crossref and DataCite; MathDB, whose entries on Carlet's question and on Conjecture 2 of [8] were still open, without solutions or comments; MathOverflow and Mathematics Stack Exchange; GitHub, including `trureturing` [13] and other repositories of AI-assisted work on open problems; Zenodo, HAL and the IACR ePrint archive; and web search. Neither [7] nor [9] had any citing work, and the only work citing [8] or [2] that we had not found on 26 September was the journal version of one we had found, on the higher weight spectra of $\mathrm{RM}_q(2,2)$. Keyword searches of Semantic Scholar again hit rate limits. We found no claim by others that 322, 326, 330 or 334 is, or is not, a weight of $\mathrm{RM}(7,14)$, and no other new work on the weight spectrum of $\mathrm{RM}(7,14)$. A last check on 9 October 2026 between 14:51 and 14:53 UTC (the arXiv, including that day's announcement in cs.IT and math.CO, where [7] and the arXiv version of [8] still had only version 1; GitHub; and MathDB) found nothing new. This reports the coverage of our searches, not a guarantee of priority.

2 Preliminaries

A *k-flat* is an affine subspace of dimension k, with 2^k points; its indicator 1_F is a product of $m-k$ affine functions, so it lies in $\mathrm{RM}(m-k,m)$. A nonempty intersection of flats is a flat. We write $x_S = \prod_{i \in S} x_i$ ($x_\emptyset = 1$). Affine bijections preserve degrees, weights and flats. We use the following known results.

w	codeword f_w	$(\sigma_1, \dots, \sigma_k)$
354	$\mathsf{a}+\mathsf{b}+x_1x_2x_3x_5x_6x_7x_{11}+x_1x_3x_6x_7x_{10}x_{12}x_{13}+x_3x_4x_6x_7x_{10}x_{12}x_{13}$	$(640, 203, 50, 12, 1)$
4378	$\mathsf{a}+\mathsf{b}+x_3x_4x_6+x_1x_5x_7x_{11}x_{13}+x_6x_{14}$	$(6912, 1537, 173, 21, 1)$
4380	$x_2x_6x_7x_9x_{12}+x_3x_4x_5x_8x_9x_{10}x_{13}+x_8x_{14}+x_1x_3x_5x_9x_{10}x_{11}x_{14}$	$(4864, 288, 25, 1)$
4382	$\mathsf{a}+\mathsf{b}+x_1x_2x_5x_6x_7+x_4x_9+x_3x_4x_6x_8x_{12}$	$(5376, 725, 150, 20, 1)$
8174	$\mathsf{a}+\mathsf{b}+x_7+x_5x_{10}+x_2x_3x_4x_5x_8x_{10}x_{11}+x_1x_2x_4x_6x_{13}x_{14}$	$(12928, 2825, 314, 59, 9, 1)$
8178	$\mathsf{a}+\mathsf{b}+x_7+x_2x_5x_6x_{13}+x_2x_5x_8x_{10}x_{13}x_{14}$	$(9728, 1081, 187, 19, 1)$
8182	$\mathsf{a}+\mathsf{b}+x_8+x_1x_2x_3x_4x_6x_{11}x_{13}+x_1x_{14}+x_2x_7x_8x_{10}x_{11}x_{12}x_{14}$	$(12800, 2765, 304, 50, 8, 1)$
8186	$\mathsf{a}+\mathsf{b}+x_{11}+x_4x_{12}+x_1x_2x_8x_{10}x_{11}x_{12}x_{13}$	$(12672, 2597, 219, 23, 1)$
8188	$x_1+x_2x_5x_6x_8x_9x_{10}x_{13}+x_2x_{14}+x_1x_4x_6x_7x_{11}x_{12}x_{14}$	$(12544, 2370, 100, 2)$
8190	$\mathsf{a}+\mathsf{b}+x_2+x_1x_7x_8$	$(10496, 1313, 82, 1)$

Table 1: The codewords of Theorem A; $\mathsf{a} = x_1 \cdots x_7$, $\mathsf{b} = x_8 \cdots x_{14}$. If $f_w = \sum_{i=1}^{k} x_{S_i}$, then $\sigma_j = \sum_{|T|=j} 2^{14-|\bigcup_{i\in T} S_i|}$ over $T \subseteq \{1, \dots, k\}$, and $\operatorname{wt} f_w = \sum_j (-2)^{j-1}\sigma_j$ (Lemma 2.1). The complements $1 + f_w$ for $w = 354, 8190, 8188, 8186, 8182, 8178, 8174$ have the weights $16030, 8194, 8196, 8198, 8202, 8206, 8210$.

(P1) If $r < m$, every codeword of $\mathrm{RM}(r, m)$ has even weight, since $\operatorname{wt} f \bmod 2$ is the coefficient of $x_1 \cdots x_m$ in f. As $\operatorname{wt}(f+1) = 2^m - \operatorname{wt} f$, the weight spectrum is invariant under $w \mapsto 2^m - w$.

(P2) For $r \geq 1$, the weights of $\mathrm{RM}(r, m)$ are divisible by $2^{\lfloor (m-1)/r \rfloor}$ (McEliece [12]). In particular the weights of $\mathrm{RM}(6, 13)$ are divisible by 4.

(P3) For $r \leq m$, the minimum distance of $\mathrm{RM}(r, m)$ is $d = 2^{m-r}$, and the codewords of weight d are the indicators of $(m-r)$-flats [10, Chapter 13].

Lemma 2.1. *For $S_1, \dots, S_k \subseteq \{1, \dots, m\}$,*

$$\operatorname{wt}\Bigl(\sum_{i=1}^{k} x_{S_i}\Bigr) = \sum_{\emptyset \neq T \subseteq \{1,\dots,k\}} (-2)^{|T|-1}\, 2^{m-|\bigcup_{i\in T} S_i|}.$$

In particular, the weight depends only on the sizes of the $2^k - 1$ Venn regions of $S_1, \dots, S_k$.

Proof. For $y_1, \dots, y_k \in \{0, 1\}$ the exclusive or of the y_i equals $\sum_{\emptyset \neq T} (-2)^{|T|-1} \prod_{i\in T} y_i$, by induction on k from $y \oplus y' = y + y' - 2yy'$. Put $y_i = x_{S_i}(x)$ and sum over $x \in \mathbb{F}_2^m$: the product over T equals 1 exactly when $x_j = 1$ for all $j \in \bigcup_{i\in T} S_i$, which happens at $2^{m-|\bigcup_{i\in T} S_i|}$ points. The size of a union is a sum of sizes of Venn regions. □

Lemma 2.2. *Let $f \colon \mathbb{F}_2^{m+1} \to \mathbb{F}_2$, and put $g(x) = f(x, 0)$ and $g'(x) = f(x, 1)$ for $x \in \mathbb{F}_2^m$. Then $f = g + x_{m+1}(g + g')$, and $f \in \mathrm{RM}(r+1, m+1)$ if and only if $g \in \mathrm{RM}(r+1, m)$ and $g + g' \in \mathrm{RM}(r, m)$. Moreover $\operatorname{wt} f = \operatorname{wt} g + \operatorname{wt} g' = \operatorname{wt}(g + g') + 2|\operatorname{supp} g \cap \operatorname{supp} g'|$, and $\operatorname{supp} g \cap \operatorname{supp} g' = \operatorname{supp} g \setminus \operatorname{supp}(g + g')$.*

Proof. The identity holds at $x_{m+1} = 0$ and at $x_{m+1} = 1$. The polynomials g and $x_{m+1}(g + g')$ have no monomial in common, so $\deg f \leq r+1$ if and only if $\deg g \leq r+1$ and $\deg(g+g') \leq r$. The rest is counting. □

3 Explicit codewords: Theorems A and B

Proof of Theorem A. Each f_w in Table 1 is a sum of distinct monomials of degree at most 7, so $f_w \in \mathrm{RM}(7, 14)$. By Lemma 2.1, $\operatorname{wt} f_w = \sum_j (-2)^{j-1}\sigma_j$ with the σ_j of the table; for instance $\operatorname{wt} f_{354} = 640 - 2 \cdot 203 + 4 \cdot 50 - 8 \cdot 12 + 16 = 354$. The other seven weights are those of the complements $1 + f_w$, by (P1); they are sums of at most seven monomials as well. □

Remark 3.1. The weights 8174 to 8210 also follow from a direct sum: if A depends only on $x_1, \dots, x_7$ and B only on $x_8, \dots, x_{14}$, then $\deg(A+B) \le 7$ and $\operatorname{wt}(A+B) = \operatorname{wt} A(128-\operatorname{wt} B)+(128-\operatorname{wt} A)\operatorname{wt} B = 8192 - 2(64-\operatorname{wt} A)(64-\operatorname{wt} B)$. As $\operatorname{wt} A$ and $\operatorname{wt} B$ can take any value from 0 to 128, taking $\operatorname{wt} B = 63$ and $\operatorname{wt} A = 64 - n$ shows that every even number $8192 - 2n$ with $|n| \le 64$ is a weight.

Proof of Theorem B. The file `certificates/rm714/data/rm714_all_witnesses.json` of version 1.4.0 of [4] (SHA-256 `784bca3b`…) lists, for each of the 7901 elements w of S_0, a sum of at most eight distinct monomials of degree at most 7 with weight w. The Lean proof checks every one of them (Section 7). □

Most entries of the file were found by random sampling of sparse polynomials, and 362 by [7, Lemma 8]; this plays no role in the proof. As S_0 consists of the weights asserted in [7, Proposition 4] and the new weights of Theorem A, the file also checks that proposition independently.

4 Construction 2 of Lou and Wang: Theorem C

Following [8], for functions f_1, f_2, f_3, f_4 on $\mathbb{F}_2^{12}$ let $f_1 \parallel f_2 \parallel f_3 \parallel f_4$ be the function on $\mathbb{F}_2^{14}$ equal to f_1, f_2, f_3, f_4 on the cosets $(x_{13}, x_{14}) = (0,0), (1,0), (0,1), (1,1)$, so that its weight is $\sum \operatorname{wt} f_i$. For $A, B \in \mathrm{RM}(6,12)$ we have $0 \parallel A \parallel B \parallel (A+B) = x_{13}A + x_{14}B \in \mathrm{RM}(7,14)$. Every codeword $g = 0 \parallel (g_1 + a_1) \parallel g_2 \parallel (g_1 + g_2 + a_2)$ of Construction 2 with $m = 6$ equals $x_{13}(g_1 + a_1) + x_{14}g_2 + (a_1 + a_2)x_{13}x_{14}$, so it lies in $\mathrm{RM}(7,14)$. If Conjecture 2 held for $m = 6$, the weights of these codewords would include 322, 326, 330, 334 and 354, which lie in its first family.

Proof of Theorem C. Let $g_1 = x_{T_0} + x_{S_1} + x_{S_2}$ and $g_2 = x_{T_0} + x_{S_3} + x_{S_4}$, where $T_0 = \{1, \dots, 6\}$ and $S_1, \dots, S_4$ are the sets of indices in the four products, so that $1 \le |S_j| \le 6$, and let $a_1, a_2 \in \mathbb{F}_2$ and $g = 0 \parallel (g_1 + a_1) \parallel g_2 \parallel (g_1 + g_2 + a_2)$, with

$$\operatorname{wt} g = \operatorname{wt}(g_1 + a_1) + \operatorname{wt} g_2 + \operatorname{wt}(g_1 + g_2 + a_2).$$

If $a_1 = 1$ and $a_2 = 0$, then $\operatorname{wt} g = (4096 - \operatorname{wt} g_1) + \operatorname{wt} g_2 + \operatorname{wt}(g_1 + g_2) \ge 4096$, as $\operatorname{wt} g_2 + \operatorname{wt}(g_1 + g_2) \ge \operatorname{wt} g_1$; similarly $\operatorname{wt} g \ge 4096$ if $a_1 = 0$ and $a_2 = 1$. So let $a_1 = a_2 = a$ and suppose that $\operatorname{wt} g = 322$. The three functions $g_1 + a$, g_2 and $g_1 + g_2 + a$ sum to 0, so each of their weights is at most the sum of the other two, hence at most 161. If $a = 1$, choose $i \in S_1$ and $j \in S_2$, and put $Q = (1+x_1)(1+x_i)(1+x_j)$. This polynomial has degree at most 3, is nonzero at the zero vector, and its support is contained in that of $g_1 + 1$. By (P3), $\operatorname{wt}(g_1 + 1) \ge \operatorname{wt} Q \ge 2^{12-3} = 512 > 161$, a contradiction. So $a = 0$, and g_1, g_2 and $g_1 + g_2 = x_{S_1} + x_{S_2} + x_{S_3} + x_{S_4}$ are sums of at most four monomials of degree at most 6 in $x_1, \dots, x_{12}$, of weight at most 161. By Lemma 2.1 the weight of such a sum depends only on the sizes of the Venn regions of its index sets, and an exhaustive check of the feasible sizes shows that every weight at most 161 of such a sum belongs to the set

$$0, 64, 96, 112, 120, 124, 126, 128, 136, 144, 148, 152, 154, 156, 158, 160$$

(this necessary inclusion is verified in Lean; Section 7). Write $a \le b \le c$ for the three weights; then $a + b + c = 322$, and $c \le 160$. The listed numbers that are 2 modulo 4 are 126, 154 and 158, and $322 \equiv 2 \pmod 4$, so an odd number of a, b, c are among them; as three of them sum to at least 378, exactly one is. But none of $322 - 126 = 196$, $322 - 154 = 168$ and $322 - 158 = 164$ is a sum of two elements of $\{0, 64, 96, 112, 120, 124, 128, 136, 144, 148, 152, 156, 160\}$. So no codeword of Construction 2 with $m = 6$ has weight 322, whereas Conjecture 2 of [8] asserts that $2^8 + 2^6 + 2 = 322$ is one. □

5 Partial non-existence results for the open weights

5.1 Sums of flats

Call two flats *transversal* if they meet in exactly one point. Two transversal flats of $\mathbb{F}_2^{14}$ of dimension at least 7 both have dimension 7: if their direction spaces are V and V', then $V \cap V' = \{0\}$, so $\dim V + \dim V' \leq 14$.

Proposition D. *Let F_1, F_2, F_3 be flats of $\mathbb{F}_2^{14}$ of dimension at least 7, not necessarily distinct (the whole space is allowed). Then none of $\operatorname{wt} 1_{F_1}$, $\operatorname{wt}(1_{F_1} + 1_{F_2})$ and $\operatorname{wt}(1_{F_1} + 1_{F_2} + 1_{F_3})$ lies in U.*

Proof. The weight of $1_{F_1}+1_{F_2}+1_{F_3}$ is $\sum |F_i| - 2\sum_{i<j} |F_i \cap F_j| + 4|F_1 \cap F_2 \cap F_3|$, and a nonempty intersection of flats has a power of 2 as its size. As $4 \mid |F_i|$, the weight is 2 modulo 4 if and only if an odd number of the three pairs are transversal. The same holds for one or two flats. All elements of U are 2 modulo 4.

For one flat there is no transversal pair. Two transversal 7-flats give the weight $128+128-2 = 254$. Three pairwise transversal 7-flats give $384 - 6 + 4|F_1 \cap F_2 \cap F_3| \in \{378, 382\}$. It remains to treat exactly one transversal pair, say $F_1 \cap F_2 = \{p\}$ with $\dim F_1 = \dim F_2 = 7$, and $\dim F_3 = d \in [7, 14]$. Translate so that $p = 0$; then $F_1 = V_1$ and $F_2 = V_2$ are linear subspaces with $V_1 \oplus V_2 = \mathbb{F}_2^{14}$. Let V_3 be the direction space of F_3. The pairs (F_1, F_3) and (F_2, F_3) are not transversal.

Case $p \in F_3$. Then $F_3 = V_3$, $|V_1 \cap V_3| = 2^a$ and $|V_2 \cap V_3| = 2^b$ with $a, b \geq 1$, and the weight is $258 + 2^d - 2^{a+1} - 2^{b+1}$. Since $(V_3 \cap V_1) \oplus (V_3 \cap V_2) \subseteq V_3$, we have $a + b \leq d$. The only solutions of $258 + 2^d - 2^{a+1} - 2^{b+1} \in U$ with $1 \leq a \leq b \leq 7$ and $7 \leq d \leq 14$ are $(w, d, a, b) = (322, 7, 4, 4)$ and $(322, 8, 5, 6)$, and both violate $a + b \leq d$.

Case $p \notin F_3$. Then the weight is $254 + 2^d - 2(\alpha + \beta)$, where $\alpha = |F_1 \cap F_3|$ and $\beta = |F_2 \cap F_3|$ are 0 or powers of 2 from 2 to 128. The only solution in U is $w = 334$, $d = 7$, $\{\alpha, \beta\} = \{8, 16\}$. Then $\dim(V_3 \cap V_1) + \dim(V_3 \cap V_2) = 3 + 4 = 7 = \dim V_3$, so $V_3 = (V_3 \cap V_1) \oplus (V_3 \cap V_2)$. Write $F_3 = c + V_3$ and $c = c_1 + c_2$ with $c_i \in V_i$. As F_3 meets F_1, $c \in V_1 + V_3 = V_1 + (V_3 \cap V_2)$, so $c_2 \in V_3 \cap V_2$; similarly $c_1 \in V_3 \cap V_1$. Hence $c \in V_3$ and $0 = p \in F_3$, a contradiction.

Both statements about solutions are finite checks; by hand they are quick, as $2^{a+1} + 2^{b+1}$ and $\alpha + \beta$ have at most two binary digits 1. □

The obstruction is dimension counting: in $\mathrm{RM}(6, 12)$ the indicators of the 6-flats spanned by $e_1, \ldots, e_6$, by $e_7, \ldots, e_{12}$ and by $e_1, e_2, e_3, e_7, e_8, e_9$ sum to a codeword of weight $162 = 2.5 \cdot 64 + 2$, whereas 322 in $\mathrm{RM}(7, 14)$ would need $a = b = 4$, against $a + b \leq 7$.

5.2 Sums of few monomials

Proposition E. *No sum of at most five monomials of degree at most 7 in $x_1, \ldots, x_{14}$ (the constant 1 allowed) has weight in U. The same holds for $f \circ \phi$ for every such sum f and every affine bijection ϕ of $\mathbb{F}_2^{14}$.*

Proof. The second statement follows from the first, as ϕ preserves weights. Let $f = \sum_{i=1}^{k} x_{S_i}$ with $k \leq 5$ and $|S_i| \leq 7$; equal monomials cancel in pairs, and the constant is $x_\emptyset$. In the formula of Lemma 2.1 with $m = 14$, the terms with $|T| \geq 3$ are divisible by 4; the term of $T = \{i\}$ is $2^{14-|S_i|}$, divisible by 2^7; and the term of $T = \{i, j\}$ is $-2 \cdot 2^{14-|S_i \cup S_j|}$, which is 2 modulo 4 exactly when $S_i \cup S_j = \{1, \ldots, 14\}$, that is, when S_i and S_j are complementary sets of size 7. Hence $\operatorname{wt} f \equiv 2N \pmod 4$, where N is the number of pairs $\{i, j\}$ for which S_i and S_j are complementary sets of size 7. All elements of U are 2 modulo 4, so if $\operatorname{wt} f \in U$, then N is odd; in particular two of the sets, say $S_1 = C$ and $S_2 = \{1, \ldots, 14\} \setminus C$, are complementary sets of

size 7, and at most three sets $S_3, \dots, S_k$ remain. Splitting each union according to C and its complement, Lemma 2.1 gives

$$\operatorname{wt} f = 254 + \sum_{\emptyset \neq T \subseteq \{3,\dots,k\}} (-2)^{|T|-1} \bigl(2^{7-p_T} - 2\bigr)\bigl(2^{7-q_T} - 2\bigr),$$

where p_T and q_T are the numbers of elements of $\bigcup_{i\in T} S_i$ in C and in its complement. The right-hand side depends only on the sizes of the Venn regions of the traces of $S_3, \dots, S_k$ on C and on its complement, and these satisfy $p_{\{i\}} + q_{\{i\}} = |S_i| \le 7$; going through all possible sizes shows that the right-hand side is never in U (this enumeration is part of the Lean proof; Section 7). □

Among these sums, the weights 2 modulo 4 in $[250, 400]$ are 254, 314, 318 and all such numbers from 338 to 398. Version 1 proved Proposition E by enumerating all vectors of Venn region sizes (n_T), $\emptyset \neq T \subseteq \{1, \dots, k\}$, with $\sum_T n_T \le 14$ and $\sum_{T \ni i} n_T \le 7$ for all i, for $k \le 5$, and reported their numbers; that enumeration is not part of version 2.

5.3 A reduction

Lemma 5.1. *Let $f \in \mathrm{RM}(7, 14)$ have support S and weight $w \equiv 2 \pmod 4$. For $a \in \mathbb{F}_2^{14} \setminus \{0\}$ let n_a be the number of pairs $\{x, x+a\}$ contained in S. Then $\sum_{a\neq 0} n_a = \binom{w}{2}$, every n_a is odd, and in coordinates in which a is the last unit vector we have $f = g + x_{14}h$ with $g \in \mathrm{RM}(7, 13)$, $h \in \mathrm{RM}(6, 13)$, $n_a = |\operatorname{supp} g \setminus \operatorname{supp} h|$ and $\operatorname{wt} h = w - 2n_a$.*

Proof. Each 2-subset $\{x, y\}$ of S is counted once, by $a = x + y$. Composing f with an invertible linear map sending e_{14} to a, we may assume $a = e_{14}$. By Lemma 2.2 with $m = 13$ and $r = 6$, $f = g + x_{14}h$ with $g = f(\cdot, 0) \in \mathrm{RM}(7, 13)$ and $h = f(\cdot, 0) + f(\cdot, 1) \in \mathrm{RM}(6, 13)$; the pairs in direction e_{14} correspond to the points of $\operatorname{supp} g \cap \operatorname{supp}(g + h) = \operatorname{supp} g \setminus \operatorname{supp} h$, and $\operatorname{wt} h = w - 2n_a$. By (P2), $4 \mid \operatorname{wt} h$, so $2n_a \equiv w \equiv 2 \pmod 4$. □

Proposition F. *Let $w \in U_0$. Then w is a weight of $\mathrm{RM}(7, 14)$ if and only if there are $h \in \mathrm{RM}(6, 13)$ and $g \in \mathrm{RM}(7, 13)$ with*

$$(\operatorname{wt} h,\ |\operatorname{supp} g \setminus \operatorname{supp} h|) = (w - 2, 1) \quad \text{or} \quad (w - 6, 3).$$

Moreover, a codeword of weight $w = 322, 326, 330, 334$ has such a decomposition, as in Lemma 5.1, in at least 7559, 7235, 6908, 6576 directions.

Proof. If g and h are as stated, then $f = g + x_{14}h \in \mathrm{RM}(7, 14)$ has weight $\operatorname{wt} h + 2|\operatorname{supp} g \setminus \operatorname{supp} h| = w$ by Lemma 2.2. Conversely, let f have weight w. By Lemma 5.1 the n_a are odd and $\sum_{a\neq 0}(n_a - 1) = \binom{w}{2} - 16383$. A direction with $n_a \ge 5$ contributes at least 4 to this sum, so at most $\lfloor (\binom{w}{2} - 16383)/4 \rfloor$ directions have $n_a \ge 5$; for $w = 322, 326, 330, 334$ this bound is 8824, 9148, 9475, 9807. In each of the remaining directions $n_a \in \{1, 3\}$, and Lemma 5.1 gives the decomposition with $\operatorname{wt} h = w - 2n_a$. □

6 Computations

In version 2 every result of this paper is proved in Lean (Section 7), and no statement depends on a computation outside Lean. Version 1 reported computations by two separately written programs (for Theorems A and B, Lemma 4.1, Propositions C′, D, E and F) and heuristic searches for codewords with weights in U_0. They are not part of version 2, and we make no claim about them here; the programs, data and logs remain in the directory `certificates/rm714` of version 1.4.0 of the repository [4]. One observation of those searches is now a theorem: in $\mathrm{RM}(6, 12)$, where $d = 64$, the numbers 162, 166, 170 and 174, which correspond to U_0 $(2.5d + 2, \dots, 2.5d + 14)$, are weights. The searches found codewords of these weights, and the Lean proof checks them.

7 Formal verification

All the results are formally verified in Lean 4 (version 4.34.1) with Mathlib [11] (version v4.34.1, commit `d13f23b7`). Version 1 of this paper was released with a formalisation of Theorem A only, the modules `RM714Weights.lean` and `RM714WeightsAudit.lean` of version 1.4.0 of [4]. For version 2 every result of the paper was formalised: each of its mathematical claims is a statement of Table 2, a definition, a step of a written proof, or an observation that is immediate from the definitions.

Statements before formal proofs. The statements were written from the released paper and reviewed independently before the formal proofs were written. On 27 September 2026 a Claude Code agent wrote a Lean file of definitions and 44 statements for version 1 of the paper, with 78 known-answer tests of its definitions; an independent Claude Code agent reviewed it the same day, between 18:23 and 19:20 UTC, and asked for corrections and additions. Following the author's decision of 27 September that claims which cannot be formalised completely are deleted, OpenAI Codex wrote the present statement file on 6 October 2026: it deletes the 15 statements that rest on the theorems of Kasami and Tokura or of Kasami, Tokura and Azumi, or on the claims deleted with them; keeps 25 statements word for word, and the definitions of Carlet's question and of Conjecture 2 for $m = 6$; strengthens one statement and keeps the first half of another; and adds nine that the review had asked for. A Codex agent that had not written the file reviewed it, the file was compiled with 98 known-answer tests, and it was frozen on 6 October 2026 at 17:13 UTC, with the SHA-256 hashes `e14b3378bfec9e32d990ca752ca041e07b64868bcffd0109ff701d85c1417abb` (main file) and `6c5b7d004bfce3b3f71865b030f950fb010a651f770e54a3aa47b3bde1e673f8` (a short file that links its notion of weight to the one of version 1). On 7 October 2026 one statement was deleted from the main file: the five numbers of vectors in the enumeration of version 1 (`PropE_counts`), whose kernel certificates were estimated to need about a week of checking (after 13 hours, 238 of the 2712 modules had been checked, and the rate was falling). The deletion follows the author's rule that claims which cannot be formalised completely are deleted, applied to this statement by the author on 7 October 2026; the other statements are unchanged, and the file was reviewed and frozen again at 19:01 UTC, with the hash `83ea2291d96864401f5147d49e3306e2401d592cc7dc7e4ac171ccda954b6d78`. The proofs were written in separate files that import the statement files and never modify them. The earliest of them was created on 6 October 2026 at 17:15 UTC, two minutes after the freeze, and the last was accepted on 8 October 2026 at 02:11 UTC; the proofs were submitted in twelve groups, each accepted after two builds from empty directories and a review of the evidence by a Codex agent that had not written it. The proof of Theorem A reuses the codeword data of version 1, given as bit masks, and checks their weights in the kernel as the Lean module of version 1 did; that module was written before the statement files (see "How the proofs work").

What is proved. The statement files contain 35 propositions, listed in Table 2; two of them restate others with Mathlib's notions only. The file of checks (`lean/Research/Backfill/Paper6/CheckPaper6.lean`, namespace `Paper6FinalPackageV4`) proves each of them, by a theorem `check_X` of type `BackfillPaper6.V3.X`, the name of the statement in the statement file.

How the proofs work. The formal proofs follow the written ones, with the following differences. *Preliminaries.* For (P1), the weight modulo 2 is written as the sum of all values of the polynomial, which is the sum of the coefficients of the monomials in which every variable occurs. (P2) is proved for every polynomial from Lemma 2.1, applied to its monomials with repetitions: each term $(-2)^{k-1}2^{m-u}$ has $u \le rk$ and $u \le m$, which gives the required power of 2. The

In the paper	Lean statement
(P1), (P2)	`P1`, `P2`
(P3) and the minimum distance	`P3_minWeight`, `MinDist`
flats and affine bijections (Section 2)	`FlatBasics`, `AffineInvariance`
Lemmas 2.1 and 2.2	`Lem21`, `Lem21_venn`, `Lem22`
Theorem A and Table 1	`ThmA`, `ThmA_mathlib`, `Table1`
the 14 elements of $M \setminus U$ (Section 1)	`IntroMU`
Remark 3.1	`Rem31`
Theorem B and $\lvert S_0\rvert = 7901$	`ThmB_witnesses`, `S0_card`
S_0 and [7, Proposition 4] (Section 3)	`S0_eq_LA`, `LAProp4`
the Corollary	`CarletEquiv`
Construction 2 for $m = 6$: codewords, and the conjectured family containing 322, 326, 330, 334 and 354	`LWConstrInRM`, `LWConj2_gives`
concatenations (Section 4)	`Sec4Concat`, `Cat4Weight`
Theorem C	`ThmC_literal`
transversal flats, Proposition D, the example 162	`TransversalDim`, `PropD`, `RemD162`
Proposition E and the remark after it	`PropE`, `PropE_mathlib`, `PropE_remark_2mod4`
Lemma 5.1 and Proposition F	`Lem51`, `PropF`
the weights 162, 166, 170 and 174 of RM$(6, 12)$ (Section 6)	`Calib612`
the notion of weight of version 1	`Sec7Bridge`
fact $(*)$ of version 1, no longer used	`FactStar`

Table 2: The 35 statements of the statement files, in the namespace `BackfillPaper6.V3`.

minimum distance (`MinDist`) is proved by induction on m with Lemma 2.2, with a codeword of weight 2^{m-r} built recursively. For (P3), if a codeword of minimum weight is nonzero at x and at $x + a$, a nonzero difference of its two halves in the direction a would violate the bound on the minimum distance, so translation by a preserves the support; after translating one point of the support to zero, the support is a linear subspace, and the indicator of the resulting flat is a product of affine forms (`FlatBasics`). Lemma 2.1 is proved by expanding the exclusive or and summing over the points, and Lemma 2.2 by splitting off the last variable. *Theorems A and B.* The codewords of Table 1 are the lists of monomials of version 1, given as bit masks. The ten rows are checked by 99 small kernel certificates, which include all 49 entries of the σ-columns together with their structural and weight data; Lemma 2.1 connects the finite sums to the actual weights. The 7901 codewords of Theorem B are checked in 65 blocks of at most 256 weights, each list of monomials being checked for its degree, its number of monomials and its weight by a kernel certificate, and the blocks are shown to cover S_0. The middle interval is counted by the injection $n \mapsto 320 + 2n$ for $0 \le n < 7873$; adding the two disjoint sets of eighteen end values and removing the eight elements of U gives $\lvert S_0\rvert = 7901$. The comparison with $M \setminus U$ and with the set of [7, Proposition 4] uses a symbolic partition of the intervals and their complements, with the remaining small finite facts checked in the kernel. *Theorem C.* The sums of at most four monomials of degree at most 6 in twelve variables are classified by the sizes of their Venn regions, and the weights at most 161 are collected into the list of sixteen values of the written proof. The cases $a_1 \neq a_2$ use the four block weights and the triangle inequality as in the written proof. In the case $a_1 = a_2 = 1$, both the written and the formal proof use a nonzero polynomial of degree at most 3 whose support lies in that of $g_1 + 1$. The minimum distance of RM$(3, 12)$ gives $\mathrm{wt}(g_1 + 1) \ge 2^9 = 512 > 161$, a contradiction. *Propositions D, E and F.* For Proposition D, the inclusion–exclusion for indicators, the sizes of flats and the constraint modulo 4 reduce to one transversal pair; the dimensions of directions and of affine spans bound the sizes of the intersections, a finite integer certificate in the kernel handles the remaining cases, and

the other transversal configurations are excluded by parity and arithmetic. For Proposition E, the necessity modulo 4 gives the complementary pair of monomials of degree 7; the remaining at most three monomials are described by the joint Venn regions of their traces on the two blocks of seven variables, the finitely many cases are checked by kernel certificates against the weight formula, and affine invariance (`AffineInvariance`) gives the second statement. For the remark after Proposition E, the nineteen weights are given by explicit sums of at most five monomials (for 362, $\mathsf{a}+\mathsf{b}+x_1x_2x_3x_4x_5x_8x_9+x_1x_2x_3x_4x_5x_{10}x_{11}+x_1x_2x_3x_4x_5x_{12}x_{13}$), and the other numbers congruent to 2 modulo 4 in $[250, 400]$ are excluded by the interval bound of the proof of Proposition E. Lemma 5.1 uses a linear change of coordinates for each direction and Lemma 2.2; the identity $\sum_{a\neq 0} n_a = \binom{w}{2}$ is an exchange of finite sums, the parity of n_a comes from (P2), and Proposition F follows as written, with the four bounds checked by arithmetic. *Other statements.* Remark 3.1 interpolates an arbitrary function of seven variables by a polynomial and decomposes the weight of a direct sum over the product of the two blocks. The weights 162, 166, 170 and 174 of $\mathrm{RM}(6, 12)$ are those of four explicit polynomials of degree 6, checked on their 4096-point truth tables in blocks. The statement `Sec7Bridge` connects the list-of-monomials predicate of version 1 to the polynomials used here in both directions, with repeated monomials cancelling.

Checks. For every statement of Table 2, the recorded axiom audits (`#print axioms`) report only the axioms `propext`, `Classical.choice` and `Quot.sound`, or none. A static scan of the code of the package, outside comments, found no `sorry`, `admit`, `native_decide`, `bv_decide`, `implemented_by`, `extern` or `unsafe`, and no `axiom` declaration. The Lean package of this paper, 375 modules (the two statement files, the tests, 370 proof modules, the file of checks, and a copy of the module `RM714Weights` of version 1, which the statement `Sec7Bridge` imports; the copy differs from the original only in the names of two library lemmas), was built module by module in two new empty directories, reusing only the pinned toolchain and the compiled Mathlib and its dependencies, and every module was checked with `leanchecker`; all 1500 commands ended with exit code 0 and printed no warning (8 October 2026, 02:13–05:37 UTC; the two builds ran side by side). The 98 known-answer tests were compiled and checked in both builds, and a Codex agent that had written none of the proofs reviewed the evidence and approved it. The package was then arranged in the directory layout of the repository: local import paths and the documentation comment of the file of checks were updated, and the compatibility copy of `RM714Weights` was replaced by the unchanged published module. The same 1500 compile and `leanchecker` commands were then run in two further new empty directories on 8 October 2026, 17:10:53–20:30:57 UTC, and all ended with exit code 0; each build again checked the 98 known-answer tests and the 35 statements. The unchanged published module `RM714Weights` printed exactly two known deprecation warnings in each build, for `if_pos` and `if_neg` (four in total); the other 374 modules and all `leanchecker` commands printed no diagnostics. A Codex agent that had written none of the proofs approved this validation on 8 October 2026 at 21:19 UTC. For the release, the whole Lean project of the repository was rebuilt from a clean directory and all its modules were checked again with `leanchecker` (9 October 2026, 00:02–03:27 UTC); as one `lake build` of the whole project takes longer than our limit of 30 minutes per command, the modules were built in dependency order, at most eight per command. The 35 theorems were also checked locally with comparator, which compares each statement with a challenge file and replays the proofs in the Lean kernel and in the independent kernel nanoda: 32 of them in a run of comparator for that theorem alone; the three statements on Proposition E (a run for `PropE` alone took 42 minutes, one for `PropE_remark_2mod4` alone was stopped at the limit, and `PropE_mathlib` alone was not run) by the same checks made one after another on one export of the three theorems, with a version of comparator modified for this purpose by Codex (not published), the Lean kernel replaying the export in 47 steps (the README of the repository describes this). The statement files, the tests, the proofs and the file of checks are in

version 1.13.0 of the repository [4], directory `lean/Research/Backfill/Paper6/`; the receipt of the two builds is in `certificates/rm714/lean-package-v2/`, and the logs of the rebuild of the whole project and of the comparator check are in `logs/`.

Definitions the reader must trust. From Mathlib the statements use `ZMod 2`, functions `Fin m → ZMod 2` as the points of $\mathbb{F}_2^m$, multivariate polynomials with their total degree, evaluation and support, the Hamming weight `hammingNorm`, supports, cardinalities, affine subspaces with their direction and dimension, and affine and linear bijections. The statement file defines $\mathrm{RM}(r, m)$ as the set of truth tables of polynomials of total degree at most r (as $x_i^2 = x_i$ on $\mathbb{F}_2$, this is the code of the paper), the weights and the weight spectrum, k-flats as nonempty affine subspaces whose direction has dimension k, indicators, the sets U_0, U, K, S_0, M and the set W_0 of [7, Proposition 4], the codewords of Table 1 and the σ-columns, direct sums, the concatenation $f_1 \,\|\, f_2 \,\|\, f_3 \,\|\, f_4$, Construction 2 with $m = 6$, our reading of Carlet's question and of Conjecture 2 for $m = 6$, the vectors of Venn region sizes of the proof of Proposition E in version 1 (now used only in the tests, as are the Kasami–Tokura normal forms and the list of weights of (P4b) of version 1), the numbers n_a of Lemma 5.1, and the decompositions of Proposition F. The 98 known-answer tests check these definitions on examples, including examples that must fail.

Faithfulness and non-claims. The set K is data: no statement says that it is the set of weights below 320. The statement `CarletEquiv` concerns our reading of Carlet's question, stated before the Corollary. The formal Theorem C concerns Construction 2 as written in [8]: it says that no codeword of it with $m = 6$ has weight 322, and that Conjecture 2 for $m = 6$, formalised with both of its families, is false. "Sums of at most k monomials of degree at most r" are polynomials with at most k terms and total degree at most r; the constant is allowed. In Lemma 5.1, "coordinates in which a is the last unit vector" means: for every linear bijection ϕ of $\mathbb{F}_2^{14}$ with $\phi(e_{14}) = a$. The statement `FactStar` is the fact $(*)$ used by Proposition C′ of version 1 (if G, E are flats of $\mathbb{F}_2^{12}$ and q has degree at most 2, then $|G \cap E \cap \operatorname{supp} q|$ is even or at most 3); it was kept and proved, although no proof of version 2 uses it. The statement `Sec7Bridge` says that the weight predicate of version 1, `RM714.IsRM714Weight`, agrees with the weights of $\mathrm{RM}(7, 14)$ defined here; it carries no bound on the number of monomials, which `ThmA` does. The enumerations in the written proofs of Theorem C and Proposition E are carried out inside the Lean proofs; the other steps of the written proofs are verified only through the results they prove.

Changes in version 2. Apart from this section, Section 6, the abstract, the disclosure and the dates, version 2 differs from version 1 as follows. The title of version 1 claimed the whole weight spectrum up to four weights; the determination of the weights below 320 rested on the theorem of Kasami, Tokura and Azumi [6], which we could not consult, and is deleted, with (P4a) and the assertion of Theorem B that the weight spectrum of $\mathrm{RM}(7, 14)$ is $S_0 \cup X$ for some $X \subseteq U$. Also deleted, because they rest on the theorems of Kasami and Tokura [5] or of Kasami, Tokura and Azumi [6]: the normal forms in (P3) and their consequences for $\mathrm{RM}(6, 12)$; (P4b), the weights of $\mathrm{RM}(6, 12)$ below 160; Lemma 4.1; the extension of Theorem C to arbitrary $g_1, g_2 \in \mathrm{RM}(6, 12)$; Proposition C′ and the remark after it; and the statement after Proposition E that the weights below 320 of those sums are exactly the elements of K. The heuristic searches of Section 6.2 and Table 2 of version 1 are not reported. (P2) now states the hypothesis $r \geq 1$, which version 1 omitted. Theorem C is attributed to Lou and Wang [9], and its written proof no longer uses (P4b). The written proof of Proposition E replaces the enumeration of version 1 by a reduction modulo 4, and the counts of that enumeration are deleted. The Corollary is stated separately. Section 8 no longer describes a route through [6] and the linear-algebra criterion of Section 6.2 of version 1, and cites Open problem 1 of [9]; the paragraph on prior work now describes [9]; the definition of a quadratic function and

the reference to Carlet's AIMS Math. paper, used only in (P3), (P4a) and Proposition C′ of version 1, are dropped. Apart from the deletions listed here, the statements of the remaining results are unchanged.

8 Open questions

Are 322, 326, 330 and 334 weights of RM(7, 14)? By the Corollary this decides Carlet's question for $(c, m) = (7, 14)$, and for 322 it is Open problem 1 of [9]. Proposition F reduces each of these questions to one about RM(6, 13) and RM(7, 13).

Tool and computational resource disclosure

Version 1 of this paper was produced on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. Claude Code agents selected the problem, found the results and their proofs, wrote the programs and the Lean formalisation of Theorem A, carried out the literature searches described in the introduction, and drafted the text.

Version 2 was produced from 27 September 2026 to 9 October 2026. On 27 September 2026 a Claude Code agent (model Claude Opus 5.5) wrote the first statement file of Section 7, and an independent Claude Code agent reviewed it. The same day we read the note [9] of Lou and Wang, and OpenAI Codex (desktop application; its main session used the model `gpt-6-astra`) checked version 1 of the paper independently: it found that the determination of the weight spectrum rested on the theorem of Kasami, Tokura and Azumi, which we had not verified, that (P2) lacked the hypothesis $r \geq 1$, and that the example after Proposition D was stated without its membership in RM(6, 12). On 6 October 2026 Codex wrote the present statement file, which a Codex agent that had not written it reviewed before it was frozen, and from 6 to 8 October 2026 Codex wrote the Lean proofs (the retained records of its main session show the model `gpt-6-astra`); each group of proofs was accepted after two builds from empty directories and a review of the evidence by a Codex agent that had not written it, and on 8 October 2026 the assembled package was built and checked in two new empty directories and reviewed in the same way. On 8 October 2026 Codex also checked the draft of version 2 paragraph by paragraph against the statement files, the file of checks and the package receipts, and on 9 October the revised passages; the text was revised accordingly. On 9 October 2026, for the release, Codex rebuilt the whole Lean project of the repository and checked the 35 theorems with comparator, and a Codex agent that had written none of the proofs or tools reviewed the evidence. Claude Code drafted and revised the text of version 2 (the session records show the model Claude Opus 5.5, and from 7 October 2026 also Claude Fable 5.1). An independent Claude Code agent (model Claude Fable 5.1), which had written neither the paper nor any Lean file, refereed the draft of version 2 on 7 October 2026 (UTC): it compared every statement of Table 2 with the frozen files, re-derived the written proofs, and re-checked the finite enumerations of Sections 4 and 5 and the 7901 codewords of Theorem B by computations of its own; its corrections were applied. Throughout, "we" refers to this workflow; the reading and searching described in the paper were done by the agents. The author is not a professional mathematician. The author has read version 2 of the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. No human mathematician has reviewed this paper. The correctness of the results rests on the Lean proofs of the statements in Table 2 and on the agreement of those statements with the paper, which was checked by the reviewing agents; the written proofs are given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

References

[1] C. Carlet, The weight spectrum of the Reed–Muller codes RM($m-5, m$), extended abstract, Discrete Mathematics Days 2024, https://dmd2024.web.uah.es/files/abstracts/paper_10.pdf.

[2] C. Carlet, The weight spectrum of the Reed–Muller codes RM($m-5, m$), *IEEE Trans. Inf. Theory* **70**(7) (2024), 4799–4807, https://doi.org/10.1109/TIT.2023.3343697.

[3] C. Carlet and P. Solé, The weight spectrum of two families of Reed–Muller codes, *Discrete Math.* **346**(10) (2023), 113568, https://doi.org/10.1016/j.disc.2023.113568.

[4] H. Dong, *ai4math-results*, repository, https://github.com/Horace-Maxwell/ai4math-results, versions 1.4.0 and 1.13.0 (2026); archived at Zenodo under the concept DOI https://doi.org/10.5281/zenodo.22970254.

[5] T. Kasami and N. Tokura, On the weight structure of Reed–Muller codes, *IEEE Trans. Inf. Theory* **16**(6) (1970), 752–759, https://doi.org/10.1109/TIT.1970.1054545.

[6] T. Kasami, N. Tokura and S. Azumi, On the weight enumeration of weights less than 2.5d of Reed–Muller codes, *Inform. and Control* **30**(4) (1976), 380–395, https://doi.org/10.1016/S0019-9958(76)90355-7.

[7] M. Leuenberger and M. Albrizzio, On the weight spectrum of the Reed–Muller codes RM(7, 14), arXiv:2606.21425v1, 19 June 2026.

[8] Y. Lou and Q. Wang, Determining the weight spectrum of the Reed–Muller codes RM($m-6, m$), *Des. Codes Cryptogr.* **93** (2025), 4925–4936, https://doi.org/10.1007/s10623-025-01708-7; quoted from arXiv:2406.03803v1.

[9] Y. Lou and Q. Wang, A note on two conjectures about the weight spectra of the Reed–Muller codes, *Discrete Appl. Math.* **388** (2026), 142–145, https://doi.org/10.1016/j.dam.2026.03.038.

[10] F. J. MacWilliams and N. J. A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland, Amsterdam, 1977.

[11] The mathlib Community, The Lean mathematical library, in *CPP 2020*, ACM, 2020, https://doi.org/10.1145/3372885.3373824.

[12] R. J. McEliece, Weight congruences for p-ary cyclic codes, *Discrete Math.* **3** (1972), 177–192.

[13] The Omega Institute, `trureturing` repository, https://github.com/the-omega-institute/trureturing, accessed 26 September 2026.